CausalCache: Budget-Conditioned Behavioral Restoration for Long-Horizon GUI Memory

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Abstract

Long-horizon GUI agents must decide which historical visual evidence to expose to an action policy under a limited multimodal context budget. Existing memory controllers commonly learn from recency, similarity, attention, or salience labels, none of which directly measures whether high-fidelity evidence changes the policy's executable decision. We introduce CausalCache, a policy-grounded supervision framework in which every memory contains the same deterministically serialized event summaries, while a high-fidelity restoration adds exactly one archived post-action image. We measure how much this visual addition restores a frozen policy's stable full-history self-behavior. We formulate a budget-conditioned, Shapley-style restoration value whose coalitions match the deployment budget, then distill the expensive offline values into a query-conditioned multi-budget gate. The versioned reference is screened for exact action-contract parsing, finite teacher-forced logits, and repeat-stable canonical output; expert top-1 alignment is reported only as a diagnostic and never used to admit or remove states. We evaluate whether restoration supervision improves the closed-loop success–memory frontier and whether retained restoration value predicts success after controlling for next-action likelihood, context cost, and task horizon. **[Pending: The v1 expert-aligned reference stack and first v2 output contract remain rejected. The preregistered v2.1 interface fixes GUI-Owl-1.5-8B-Instruct as a stable self-behavior reference; insert no method-effect claim until the substrate and untouched restoration gates pass.]**

Introduction

A GUI trajectory can contain screenshots, actions, text arguments, coordinates, and state changes accumulated over dozens of decisions. Passing every event to a multimodal policy is expensive and can amplify context rot, while replacing all screenshots with summaries discards exact visual evidence. A memory controller must therefore decide which historical events deserve high-fidelity exposure at each decision.

Prior controllers select memory using recency, retrieval similarity, attention, learned salience, or task-state heuristics. These signals need not agree with the frozen action policy's actual dependence on an event. Two memories can even have similar successor-action negative log-likelihood (NLL) while supporting different closed-loop outcomes. This motivates a narrower question: *how much does adding one event's archived post-action image to an otherwise unchanged strong low-fidelity history restore the frozen policy's stable full-history action behavior?*

CausalCacheanswers this question offline and uses the answers as supervision. It maintains a raw event archive, a cheap summary/index, and a limited policy-visible high-fidelity context. For preregistered stable-reference states, it samples near-budget mixed-fidelity coalitions and estimates each event's marginal behavioral restoration. All events retain byte-identical low-fidelity summaries in every coalition; selecting an event adds only its post-action image. The estimator is explicitly Shapley-style; our claimed contribution is the GUI intervention surface, the policy-behavior value, its alignment with the deployment budget, and closed-loop validation, not a new generic attribution estimator. A lightweight query-time gate then predicts these values from the current state, event summary, optional cheap visual embedding, age, and requested budget.

Our evaluation is organized around four falsifiable questions:

1. Does a restoration oracle improve executable task success at a fixed policy-visible visual budget?
2. Can a distilled gate approach the oracle without material online latency?
3. Does restoration supervision outperform the same gate architecture trained on next-action NLL or shuffled labels?
4. After matching or controlling for NLL, tokens, and horizon, does retained restoration value still explain terminal success?

Related Work

GUI memory. MementoGUI learns multimodal write, compression, and retrieval operators for a frozen GUI action model (Zeng et al. 2026). AndroTMem retrieves anchored intermediate states connected to the current subgoal (Shi et al. 2026). These systems establish the importance of learned controllers and dependency-critical events. CausalCacheinstead derives selector supervision from the marginal behavior change induced by restoring the high-fidelity version of the same event.

Interventional memory selection. CMI compares agent behavior under memory interventions (Srivastava 2026), while ATMem estimates how active memory states affect GUI task completion (Liu et al. 2026). Our scope is intentionally narrower: we hold the action policy fixed, intervene on low- versus high-fidelity GUI evidence, assign credit near the actual context budget, and validate the dense decision-time surrogate through executable closed-loop outcomes.

Shapley-style attribution. Permutation marginal contributions originate in cooperative game theory (Shapley 1953). We use this tool as an estimator, but change its value function and coalition distribution to reflect mixed-fidelity GUI context and a fixed operating budget.

Problem Formulation

Let x_t be the current observation at decision t, and let its history be

$$H_t = \{e_1, \ldots, e_{t-1}\}, \qquad e_j = (x_j, a_j, x_{j+1}, \delta_j), \quad (1)$$

where δ_j is an optional deterministic UI delta. Every event has an archived high-fidelity visual increment and a deterministic low-fidelity record $\tilde{e}_j$. We exclude a redundant candidate when its post-state is already the current observation:

$$C_t = \{j < t : \mathrm{post}(e_j) \not\equiv x_t\}. \quad (2)$$

A query-time selector chooses which candidate images to expose:

$$M_t \subseteq C_t, \qquad \sum_{j \in M_t} c_j \leq B, \quad (3)$$

where c_j is the visual-token cost and B is the policy-visible context budget. Every event, including events outside C_t, contributes the same low-fidelity record in every memory. Selecting $j \in C_t$ adds its post-action image and nothing else. Raw events are retained in an external archive, so an event can be retrieved again at a later query; B is not a persistent storage or eviction budget.

The action policy π_0 is frozen. Only the selector is trained. Our goal is to improve executable task success across budgets without attributing gains to extra visual tokens or a stronger action policy.

CausalCache

Two-Level Event Representation

The deterministic low-fidelity record contains

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step_id, action_type, action_argument,
foreground_app,
screen_text_added, screen_text_removed,
screen_change,
executor_result.
```

The record is present for every event in every coalition and its serialization is byte-identical between low- and high-fidelity conditions. Accessibility content, or a pinned OCR fallback, supplies the normalized screen-text delta. For a selected event, the policy-visible high-fidelity increment is exactly one post-action image: no before image, additional action text, or raw-argument text is added. The raw archive may retain both screenshots and execution metadata, but those fields are not part of the restoration intervention. We report the actual policy-visible text tokens rather than assuming exactly fixed text cost. The gate may additionally use a precomputed cheap visual embedding z_j; this avoids asking the summary alone to predict the importance of information that the summary necessarily omitted. The action policy receives mixed-fidelity inputs through a normal forward pass; we do not splice cached keys and values across incompatible contexts.

Stable Full-History Self-Behavior Reference

Let $X_t(S)$ denote the shared low-fidelity history with post-action images added only for $S \subseteq C_t$. The frozen-policy reference is

$$q_t = \pi_0(\cdot \mid g, x_t, X_t(C_t)). \quad (4)$$

Its deterministically decoded output must be one exact native MobileUse tool call and is used as the canonical action a_t^*. The substrate requires contract-parseable output, finite teacher-forced logits, and identical canonical actions across two deterministic forwards. Expert agreement does not define or filter the reference. Once the untouched confirm split is opened, parse or stability failures count against the fixed-denominator gate rather than being removed. We use the checkpoint's official tool-template path rather than reproducing its schema in a hand-written prompt. For teacher forcing, the processor's native assistant prefix is followed directly by the deterministic MobileUse tool call; no generated action description or fixed Action carrier is inserted. Text arguments must already use Unicode NFKC normalization.

Executable Behavioral Distance

For a coalition S, all events retain their low-fidelity records, and only events in S add images. We measure pathwise divergence along the canonical action rather than claiming a full sequence-action KL:

$$D_t(S) = \frac{1}{L} \sum_{l=1}^{L} \mathrm{KL}\big(q_{t,l}(\cdot \mid a^*_{t,<l}) \,\|\, q^S_{t,l}(\cdot \mid a^*_{t,<l})\big). \quad (5)$$

The KL is teacher-forced over the full vocabulary from the opening <tool_call> through the model-emitted closing tag. Standard model EOS tokens that are suppressed during generation receive the same finite minimum logit mask in both reference and candidate distributions before normalization. Action-type match, canonical coordinate-bin match, normalized coordinate distance, and text-argument match are reported as diagnostics, not mixed into the primary restoration distance.

Budget-Conditioned Restoration

For event j, let $\mathcal{C}_B(j)$ be a distribution over feasible coalitions $S \subseteq C_t \setminus \{j\}$ that leave room for c_j and are near budget. The budget-conditioned restoration value is

$$G^{(B)}_{j,t} = \mathbb{E}_{S \sim \mathcal{C}_B(j)} \left[D_t(S) - D_t(S \cup \{j\}) \right]. \quad (6)$$

Algorithm 1 Offline Budget-Conditioned Restoration Labels

Require: Stable-reference state $(g, x_t, H_t, C_t, a_t^*)$, budget B, samples K
Include the shared low-fidelity record for every event
Run $X_t(C_t)$ once to cache full-distinct-history behavior q_t
for $k = 1$ to K **do**
  Sample a shared near-budget coalition design for all events
  for each event j evaluated by the design **do**
    Run the policy on $X_t(S)$ and $X_t(S \cup \{j\})$
    Record $D_t(S) - D_t(S \cup \{j\})$
  end for
end for
Return means, standard errors, and sample metadata

When all high-fidelity blocks have equal cost c and B admits $b = \lfloor B/c \rfloor$ blocks, $\mathcal{C}_B(j)$ is uniform over coalitions of size $b-1$. Unlike a standard Shapley value averaged over all coalition sizes, Equation 6 asks whether restoring j helps when the system already operates near its deployed budget. We use shared, antithetic, or stratified coalition samples where applicable and report standard errors, ranking stability, and top-B overlap across seeds.

Multi-Budget Query-Time Gate

The distilled selector predicts

$$s_{j,t} = f_\theta(g, o_t, \tilde{e}_j, z_j, \Delta t, B) \tag{7}$$

with regression, pairwise-ranking, and budget-aware set losses. At inference, the selector solves a cost-constrained positive-value selection problem:

$$M_t = \arg \max_{M : \sum_{j \in M} c_j \le B} \sum_{j \in M} \max(s_{j,t} - \tau, 0). \tag{8}$$

For equal-cost event blocks this reduces to selecting up to, rather than exactly, the top $b = \lfloor B/c \rfloor$ events above threshold τ. This permits a null decision when all candidate restorations are predicted to be harmful or useless.

Experimental Protocol

Benchmarks and Policies

AndroidWorld (Rawles et al. 2025) is the sole primary closed-loop benchmark. GUIOdyssey (Lu et al. 2025) or an equivalent long-horizon mobile dataset provides offline attribution and action analysis. We use one primary frozen policy and one transfer backbone. Every comparison shares the same low-fidelity channel, action policy, gate architecture where applicable, and visual-token accounting.

The v2.1 primary policy is frozen to GUI-Owl-1.5-8B-Instruct (Xu et al. 2026) at model revision `06d5faec`, using bfloat16 and deterministic decoding through its official tool template. The earlier GUI-Owl output-contract and UI-TARS expert-alignment gates remain valid negative results; v2.1 versions the interface rather than relabeling those outputs.

The untouched confirm set contains one decision-6 state from each of 20 trajectories selected before policy inference in frozen hash order. Every state contains summaries for events 1–5. Events 1–4 are the four visual candidates; event 5 is summary-only because its post-state is the current observation. Consequently the reference contains four historical post-state images plus the current observation, covering every non-current historical event once without duplicating event 5's current-equivalent image. The primary deployment budget permits up to two of the four candidates; an exact-two cardinality ablation separates abstention from ranking quality. No state is filtered or replaced after observing parseability, stability, expert alignment, or restoration sensitivity. Image preprocessing targets 2,560 effective visual tokens per image; the actual image grid, visual tokens, and policy-visible text tokens are recorded for every run.

Reference quality is reported on two independent axes: rollout outcome where available and expert alignment. We always report action-type match, coordinate bin match, normalized coordinate distance, and text-argument match. Semantic expert match is used only when the UI target was annotated before policy output; neither axis is an admission criterion.

Comparisons

The compact main comparison includes summary only, recent, uniform-random exact expectation, OCR-plus-RGB similarity, frozen-policy vision-embedding similarity, the same gate trained on next-action NLL, the strongest reproducible GUI memory controller, a restoration oracle, and distilled CausalCache. A random-label shuffle uses the same training states and gate capacity as a negative control.

Success–Memory Frontier

We report closed-loop task success against visual tokens and end-to-end latency, plus success per unit memory/compute. Restoration must improve the Pareto frontier rather than consume additional context. **[Pending: Insert main frontier plot and paired confidence intervals.]**

Matched-NLL Mechanism Test

Within task, budget, and horizon strata, we match memory decisions by successor NLL and test whether retained restoration mass remains associated with terminal success. The preregistered conditional model is

$$\begin{aligned}\Pr(y=1) = \sigma\big(&\beta_0 + \beta_1 R + \beta_2 \mathrm{NLL} \\ &+ \beta_3 \mathrm{tokens} + \beta_4 \mathrm{horizon}\big),\end{aligned} \tag{9}$$

where R is retained restoration mass. The mechanism claim requires a stable $\beta_1 > 0$ under paired bootstrap confidence intervals and reasonable matching tolerances. We will report all eligible pairs rather than selected anecdotes.

Attribution Cost and Stability

We vary coalition samples K and report attribution standard error, Spearman rank correlation to a high-K estimate, top-budget Jaccard overlap, oracle success, total offline labeling compute, and online selector latency. The per-

permutation telescoping identity is used only as an implementation sanity check. Coalition reconstruction error measures non-additive interactions. Development uses $K = 4$, untouched confirmation uses $K = 16$, and the stability audit uses $K = 32$. Spearman, top-budget Jaccard, and sampled-to-exact-selector utility ratio are aggregated by the median over memory-sensitive confirm states, with deterministic tie and degenerate-state rules. Memory sensitivity is defined relative to $\epsilon = \max(10^{-4}, 10\bar{D}_{\mathrm{repeat}})$. Per-state normalized recovery is $(D_t(\varnothing) - D_t(S))/\max(D_t(\varnothing), \epsilon)$ and is averaged over the fixed 20-state confirm denominator.

Limitations and Claim Boundary

Restoration value is a dense decision-time surrogate. It is not a direct long-horizon causal effect, and its relevance to long-horizon control must be established through closed-loop success and the matched-NLL analysis. The causal language in this paper refers only to a controlled low-/high-fidelity intervention on a frozen policy's behavior. We do not identify environment-level structural causality and do not learn a world model. The method should be weakened or abandoned if restoration fails to beat recency, similarity, and attention; if its matched-NLL association disappears; if the distilled gate cannot approach the oracle; or if gains come from extra tokens. A parseable and repeat-stable self-behavior reference can still be consistently wrong with respect to an expert or the environment. Restoration therefore measures fidelity to a frozen policy, not action correctness. Expert alignment and executable closed-loop success remain separate diagnostics and validation requirements.

Conclusion

CausalCacheturns mixed-fidelity behavioral restoration into policy-grounded supervision for GUI memory selection. Its central empirical burden is not that attribution can fit a selector, but that budget-aligned restoration identifies high-fidelity evidence that improves executable long-horizon control beyond what next-action likelihood and context size explain.

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